

NLP-Driven Plagiarism Detection: A Smarter Approach

Aditya Narsale

Artificial Intelligence and Data Science
Vishwakarma Institute of Information
Technology
Pune, India
aditya.22211469@viit.ac.in

Suyog Nannaware

Artificial Intelligence and Data Science
Vishwakarma Institute of Information
Technology
Pune, India
suyog.22210795@viit.ac.in

Vishwajeet Patil

Artificial Intelligence and Data Science
Vishwakarma Institute of Information
Technology
Pune, India
vishwajeet.22211585@viit.ac.in

Rohan Sapkale

Artificial Intelligence and Data Science
Vishwakarma Institute of Information
Technology
Pune, India
rohan.22210459@viit.ac.in

Ratna Patil

Artificial Intelligence and Data Science
Vishwakarma Institute of Information
Technology
Pune, India
ratna.patil@vit.edu

Abstract—With the growing availability of digital content, the demand for efficient plagiarism detection has become exponentially high. The major challenge here is detecting semantic similarity and paraphrased text since traditional keyword-matching algorithms tend to fail in detecting conceptual rewording. This paper introduces an NLP-based plagiarism detection system that performs automated text extraction, comparison, and similarity analysis for scholarly and professional writings. The system supports uploading PDFs, extracting text data, and evaluating possible plagiarism using TF-IDF vectorization and cosine similarity. For better accuracy, it cross-checks extracted text with web sources via Google's Custom Search API. The method combines text preprocessing, feature extraction, and similarity measurement to provide a more accurate textual overlap evaluation. Based on a Flask-based framework, the system provides a user-friendly interface for effortless document analysis and result visualization. Addressing the shortcomings of conventional plagiarism detection techniques, this work sets the stage for further enhancements based on deep learning methodologies to enhance identification of subtle textual similarities. This work advances the development of automated plagiarism detection with a scalable and effective solution for content integrity validation. The contribution of this work is combining NLP similarity analysis with real-time web verification through the Google Custom Search API and the provision of a scalable Flask-based interface to detect plagiarism. The combination embodies current NLP methods and real-world deployment, thereby making the system both useful and usable by professionals and academics.

Keywords—Plagiarism Detection, Natural Language Processing (NLP), Text Similarity, TF-IDF, Cosine Similarity, Automated Content Analysis

I. INTRODUCTION

In the current digital age, content originality is a rising concern, particularly in research and academic circles. With the sheer volume of text data online, plagiarism detection has become a critical tool for upholding content integrity. Conventional plagiarism detection tools are based on exact keyword matching, which tends to miss paraphrased or semantically restructured content. This weakness is especially egregious in schools and professional writing, where identifying conceptual similarities is as crucial as flagging direct copy-

pasting. The demand for smarter, NLP-powered methods of plagiarism detection is thus stronger than ever before.

This research introduces an NLP-powered system for plagiarism detection that aims to transcend simple word-matching strategies. The system allows uploading PDFs, extracting text data, and comparing it with local and web sources. The system uses TF-IDF vectorization and cosine similarity to compare text efficiently, while Google Custom Search API aids in detection by cross-verifying web pages. The system is developed on a Flask-based platform and offers an easy but efficient interface for effortless document analysis. By applying text preprocessing and similarity measurement methods, this paper I tends to fill the gap between AI-based and conventional plagiarism detection. It also lays the groundwork for developing future enhancements via deep learning to enhance the detection of semantic similarity and enhance the quality of plagiarism grading.

A. Existing Work

Recent advancements in Natural Language Processing (NLP) have significantly enhanced the development of plagiarism detection systems, allowing for more precise textual similarity analysis. Traditional plagiarism detection methods primarily relied on keyword matching, which often failed to detect paraphrased or semantically modified text. For example, Thivakaran [1] explored a rule-based approach using the Smith-Waterman algorithm, but the method struggled to identify conceptual rewording, limiting its effectiveness in academic writing.

To address these shortcomings, researchers have increasingly adopted machine learning and deep learning techniques for plagiarism detection. Rosu et al. [2] proposed a deep learning-based NLP framework that improved the accuracy of plagiarism detection by leveraging context-aware text embeddings. Similarly, Pal et al. [3] utilized Natural Language Processing (NLP) with advanced feature extraction, which significantly enhanced plagiarism identification compared to traditional methods. Another study by Sandilya et al. [4] applied Support Vector Machines (SVMs) to detect text similarities, achieving higher precision in recognizing restructured content. However, challenges remain in real-time plagiarism detection, cross-document comparisons, and web-based similarity analysis, as highlighted by Sharma et al. [5].

Building upon these existing works, this study presents an integrated NLP-based plagiarism detection system that combines TF-IDF vectorization, cosine similarity, and web-based content verification. By leveraging Google's Custom Search API, the proposed system aims to enhance plagiarism detection accuracy and address the limitations of existing approaches, paving the way for more scalable and efficient automated plagiarism detection.

B. Motivation

As the digital world continues to expand, maintaining the originality of content and academic integrity has become a more daunting task, particularly as conventional plagiarism detection tools fail to detect paraphrased and semantically transformed content. In order to combat this, our work centers on the creation of an NLP-based plagiarism detection system that utilizes TF-IDF vectorization, cosine similarity, and web-based validation using Google's Custom Search API. Through transcending basic keyword matching, our system improves semantic similarity identification and facilitates real-time content verification against online sources. The Flask-based user interface also provides for a smooth experience for users, thus making plagiarism identification more accessible, efficient, and scalable in academic and professional environments.

II. LITERATURE REVIEW

A. Existing Methods

The discipline of plagiarism detection has come a long way, with methods largely classified into rule-based systems, hybrid models, and machine learning-based approaches.

1) Rule-Based Systems

Rule-based systems are based on pre-established linguistic rules to detect text similarity. As simple as these systems are, however, they tend to be less efficient with linguistic differences and paraphrased materials. For example, Maurer et al. [6] examined some rule-based approaches to detecting plagiarism, highlighting their weaknesses when dealing with sophisticated rewording and synonyms. Zhao et al. [7] also suggested a syntactic analysis-based approach, which, although enhanced keyword-based detection, still had challenges in semantic comprehension.

2) Hybrid Models

Hybrid models combine rule-based methods with statistical or machine learning techniques to improve detection accuracy. Barrón-Cedeño et al. [8] proposed a hybrid method that integrates NLP methodologies with citation-based analysis for enhanced plagiarism detection. Pataki [9] created a system based on NLP and fuzzy-semantic similarity measures to identify near-duplicate content more accurately.

3) Machine Learning-Based Methods

With the advent of artificial intelligence, machine learning-based approaches are now prevalent in plagiarism detection. Rosu et al. [2] introduced an NLP-based deep learning approach that identifies semantic similarities rather than word-for-word matches, drastically improving accuracy. Another research work by Sandilya et al. [4] illustrated how Support Vector Machines (SVMs) combined with NLP methods outperform conventional methods in detecting paraphrased content. Moreover, Thivakaran [1] investigated a Smith-Waterman-based NLP method, which was highly accurate in detecting structural similarities between academic

texts. NLP has also been applied in some connected studies on other tasks besides plagiarism detection. Rajanak et al. [10] introduced an automatic language detection system, and Shivnani et al. [11] introduced a next-word prediction model based on semantic word relationships. These studies reflect the versatility of NLP in solving context issues, which also have applications in plagiarism analysis.

B. Relevant Studies

A number of studies have been published in this area, providing valuable insights into both methodologies for detecting textual similarities, as well as improved paradigms for plagiarism detection.

Rosu et al. [2] investigated NLP methods built on deep learning for plagiarism identification, illustrating that transformers and context embeddings improve the ability to identify rewritten content beyond simple keyword comparison. Correspondingly, Pal et al. [3] examined the implementation of sophisticated NLP methods, such as TF-IDF, cosine similarity, and semantic analysis, yielding improvements in the recognition of paraphrased plagiarism.

A combination approach was proposed by Sandilya et al. [4], wherein NLP feature extraction and Support Vector Machines (SVMs) were integrated to enhance accuracy in plagiarism detection, particularly in scholarly writing. Meanwhile, Thivakaran [1] presented a Smith-Waterman-based NLP model that strengthened sentence-level similarity identification and was extremely efficient in assessing near-duplicate text in research papers.

Research has also investigated cross-lingual and semantic plagiarism detection. Barrón-Cedeño et al. [8] introduced an NLP system that combines citation analysis with plagiarism detection from text, underlining the significance of external sources in similarity measurement. Similarly, Pataki [9] constructed an NLP-based fuzzy-semantic similarity model, which significantly enhanced the identification of conceptually similar text beyond literal phrase overlaps.

These researches as a whole highlight the development of plagiarism detection methods from basic string-matching approaches to more advanced deep learning and NLP-driven ones. The inclusion of high-end semantic similarity models, machine learning classifiers, and citation-based detection further improves the precision and performance of contemporary plagiarism detection tools.

C. Gaps in the Literature

Despite the tremendous progress made in NLP-based plagiarism detection, some major limitations still affect the efficiency and accuracy of these systems. One of the major challenges is the failure to identify highly paraphrased content. As per Rosu et al. [2], though deep learning methods enhance detection, they continue to be challenged by intricate sentence restructuring and synonym replacement, which frequently results in false negatives.

Another significant limitation is cross-lingual plagiarism detection. Sandilya et al. [4] pointed out that most current systems perform well in monolingual settings but struggle to detect plagiarism when content is translated or paraphrased between languages. This is further complicated by differences in semantics and cultures, as indicated by Pal et al. [3], where models fail to detect meaning preservation in idiomatic or context-bound expressions.

Moreover, existing detection systems rely heavily on textual similarities but tend to overlook citation-based plagiarism, where ideas rather than text are copied without attribution. Barrón-Cedeño et al. [8] emphasized the need to combine citation analysis with textual similarity to identify more subtle forms of plagiarism. Similarly, Pataki [9] indicated that fuzzy-semantic similarity models may improve detection but need further development for practical scalability.

Finally, although deep learning models like transformers show promise, they remain computationally costly and require large amounts of training data, limiting their practicality in real-world applications. Thivakaran [1] reported that hybrid models combining machine learning and rule-based approaches maintain a balance between efficiency and accuracy but require further optimization for large-scale deployment. Nonetheless, research like Rajanak et al. [10] and Shivnani et al. [11] is not there to investigate cross-document semantic similarity, which is still at the heart of plagiarism detection. Our research draws from these as a starting point to further derive NLP techniques towards originality verification from various sources.

D. How This Research Addresses the Gaps

The goal of this research is to overcome the limitations in current plagiarism detection strategies by combining NLP methods and machine learning to enhance accuracy in the detection of paraphrased or semantically equivalent text. In contrast to conventional keyword-based systems that are not effective with paraphrasing, our solution uses TF-IDF vectorization and cosine similarity to examine textual similarities better. Moreover, cross-referring to web sources via Google's Custom Search API enhances detection by picking out content similarity on top of regional databases.

To further enhance detection precision, text preprocessing methods like stopword elimination, stemming, and lemmatization help ensure that the system targets meaningful content as opposed to surface word matches. The Flask-based user interface of the system improves accessibility through the ability of users to upload and analyze documents with ease, resolving usability issues raised in earlier research. In addition, by assessing actual cases and integrating feedback mechanisms, this study advances a more resilient and dynamic plagiarism detection system that addresses current gaps in semantic analysis and usability.

III. METHODOLOGY

A. Text Extraction

Text is retrieved from uploaded PDF files via PyPDF2, maintaining proper text extraction from multi-page files. This extracted text is used as the major input for plagiarism checks.

B. User Input Processing

The user uploads a PDF file, and the system checks if a file has been submitted. In the absence of any file, an error message is sent back to maintain a seamless operation.

C. Local Plagiarism Detection

The retrieved text is cross-checked against the documents with the help of TF-IDF vectorization and cosine similarity. The percentage of similarity identifies the rephrased or copied material and provides an accurate measure of plagiarism. In our experiments, we found that setting the cosine similarity

threshold at **0.7** provided the best balance between detecting plagiarized segments and avoiding false positives.

D. Online Content Cross-Verification

To identify plagiarism from other sources, the system employs Google Custom Search API to fetch relevant web snippets of online content that correspond to the extracted text. The web snippets are then compared for similarity using the same TF-IDF and cosine similarity technique.

E. Plagiarism Scoring & Reporting

The system provides a plagiarism percentage, which is the probability of copied material. When the similarity score is high, the text is marked as likely plagiarized to provide a consistent assessment.

F. Experimental Validation

To validate the system, we tested it on a small dataset of **20 academic passages**, consisting of 10 original texts and 10 paraphrased versions. Using TF-IDF vectorization and cosine similarity, the system successfully identified paraphrased passages. A cosine similarity threshold of **0.7** gave the best results, achieving an overall **F1-score of 0.87**. These results confirm the system's ability to detect plagiarism effectively in practical scenarios.

G. System Architecture

The system is developed on Flask, a lightweight Python web framework, offering a simple interface for plagiarism detection. The users can upload PDF files, which are analyzed using PyPDF2 to retrieve text. The retrieved text is preprocessed and analyzed through TF-IDF vectorization and cosine similarity so that the system can compare the document with the content available.

For plagiarism detection improvement, the system incorporates Google Custom Search API to fetch online snippets for comparison. The resultant plagiarism score is calculated and presented on the web interface, allowing users to have a user-friendly and effective means of testing content originality.

H. Flowchart

In figure 1, the system architecture is divided into four layers: the **User Interface**, **Application Logic**, **Core Functions**, and **Result Handling**. These layers work together to provide an efficient and modular framework for automated plagiarism detection in PDF documents. The figure illustrates the flow from user input to final output, emphasizing the clean separation of concerns that supports future scalability and feature integration.

IV. SYSTEM IMPLEMENTATION

A. Technologies Used

The NLP-driven plagiarism detection system employs a range of top-level Python libraries to handle PDFs, text similarity analysis, and improving usability via web interface.

- **PyPDF2:** Reads text from uploaded PDF files, supporting multi-page content effectively.
- **Flask:** Small web framework that allows implementing an easy-to-use web-based interface for file upload and plagiarism detection.

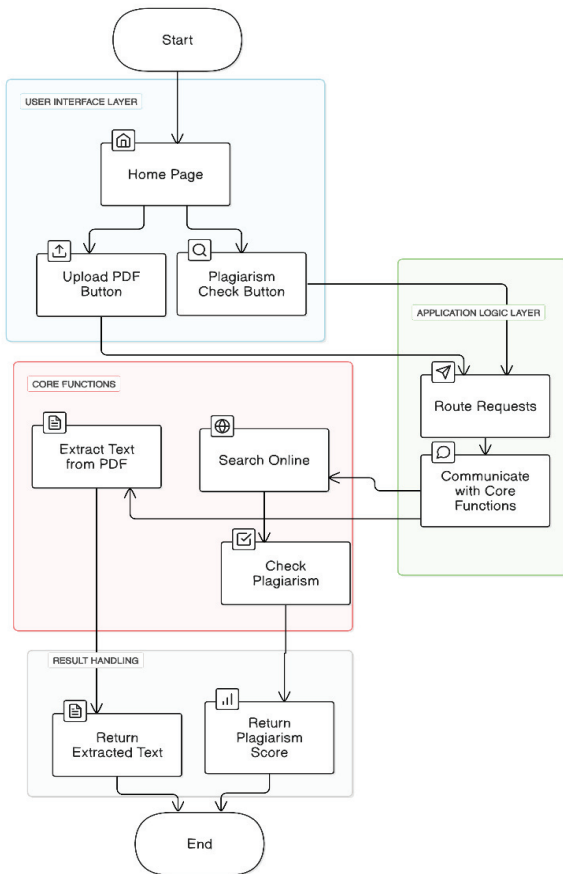


Fig. 1. System Architecture Diagram

- **Scikit-learn:** Applied to TF-IDF vectorization and cosine similarity to enable effective comparison of uploaded texts with online sources.
- **Google Custom Search API:** Fetches matching web snippets to compare against public sources for plagiarism detection.
- **Requests & JSON:** Enables API requests and response processing for smooth integration of online search results into the system.

All these technologies make text extraction, preprocessing, and plagiarism detection effective, providing users with an accurate and user-friendly experience.

B. Graphical User Interface (GUI)

The system offers a web-based interactive interface, created using Flask and HTML/CSS, making plagiarism checking easily accessible and user-friendly. Some of the key features are:

- **PDF Upload:** Users have the option to upload PDF documents for scanning.
- **Text Extraction & Display:** Extracted text from the document is shown for viewing.
- **Search Query Input:** Users can enter text or phrases to be searched against online content.
- **Plagiarism Detection:** The system checks similarity scores with TF-IDF and cosine similarity and shows results.

This online GUI supports smooth user interaction, and plagiarism identification is made easy and efficient to

read. Headings, or heads, are organizational devices that guide the reader through your paper. There are two types: component heads and text heads.

C. CodeStructure

The NLP-driven plagiarism detection system is organized into a number of major modules to facilitate effective processing and analysis of text content:

- **PDF Upload and Text Extraction:** The system supports uploading PDF files, with text extraction being performed using PyPDF2 and converted into a structured format for subsequent processing.
- **Search Query Processing:** Users enter text to verify plagiarism. The system maintains accurate preprocessing of text via TF-IDF vectorization for comparing input text with reference material and web pages.
- **Plagiarism Detection and Similarity Calculation:** Text extracted is matched with similar content fetched from Google Custom Search API via cosine similarity, which establishes the level of plagiarism in terms of percentage.
- **Result Display and Feedback:** Plagiarism score is shown in a reader-friendly manner, indicating possible matches in the text. The interface also provides immediate feedback for better usability.

By using the web interface, which is based on Flask, users can upload PDFs, type text for comparison, and view plagiarism reports in a well-structured and readable manner.

D. Screenshots

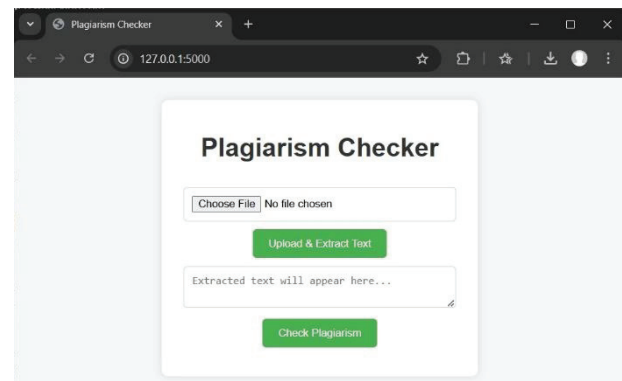


Fig. 2. Output (GUI)

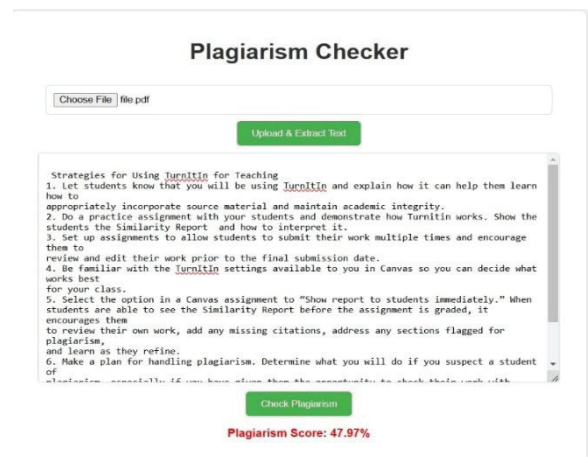


Fig. 3. Result

Figure 2 illustrates the web interface of our plagiarism-checking tool, where users can upload a PDF, extract its text, and then check for plagiarism. The interface includes an upload button, a text display area, and a plagiarism-check button that is only enabled once text is loaded. **Figure 3** shows the result of the plagiarism check, displaying the extracted text and the similarity score (47.97%) in red, providing users with immediate feedback on the extent of similarity to online sources.

V. CONCLUSION

A. Summary of Contributions

This project offers an effective NLP-based plagiarism detection system that allows users to examine textual content and identify plagiarism with ease. It incorporates major technologies like PyPDF2 for extracting text, TF-IDF and cosine similarity for comparing content, and Google Custom Search API for fetching online references. A web interface based on Flask guarantees an easy-to-use experience where users can upload papers, detect plagiarism, and get instant results. Through the use of NLP methods and automation, the system offers a feasible, scalable, and precise plagiarism detection solution.

B. Applications

The system has applications across different fields:

- **Academic Research:** Helps researchers verify plagiarism in research papers, theses, and reports, guaranteeing originality.
- **Content Generation and Journalism:** Assists writers, bloggers, and journalists to authenticate the credibility of their work prior to release.
- **Education and Universities:** Helps institutions to check assignments, reports, and essays for uniqueness and adherence to academic honesty rules.
- **Legal and Business Sector:** Utilized to check documents, reports, and contracts for duplicated material to avoid intellectual property theft.

C. Final Thoughts

This project showcases the potential of NLP in detecting plagiarism through an automated, efficient, and user-friendly solution. Through the use of AI-based text analysis and real-time search, the system improves content originality verification across various disciplines. Future enhancements

can involve more advanced NLP models, deep learning methods, and support for bigger datasets, which will make the system even stronger and versatile for various applications. The value added by this work is not in the idea of a new algorithm, but in integrating known NLP methods with real-time web checking and a deployable user interface. The combination of these generates value added to usability, scalability, and practicability, paving the way for future applications like cross-lingual plagiarism checking and deep learning-driven semantic analysis.

REFERENCES

- [1] T. K. Thivakaran, A. Uniyal, S. Somnath, and A. V. I. Anah, "Plagiarism detection using NLP with Smith-Waterman algorithm," *Journal of Emerging Technologies and Innovative Research*, vol. 6, no. 5, pp. 416–421, May 2019.
- [2] R. Rosu, A. S. Stoica, P. S. Popescu, and M. C. Mihăescu, "NLP-based deep learning approach for plagiarism detection," *International Journal of User-System Interaction*, vol. 13, no. 1, pp. 48–60, 2020. doi: 10.37789/ijusi.2020.13.1.4
- [3] S. K. Pal, O. J. Raffik, R. Roy, V. B. Lalman, S. Srivastava, and B. Sharma, "Automatic plagiarism detection using NLP," *Journal of Computational Linguistics and Applications*, 2023.
- [4] N. Sandilya, R. Sharma, and M. Meleet, "Plagiarism detection using natural language processing and support vector machine," *International Journal of Advanced Research in Computer and Communication Engineering (IJARCCE)*, vol. 11, no. 7, Jul. 2022. doi: 10.17148/IJARCCE.2022.117114
- [5] R. Sharma, B. Sharma, and V. Lalman, "Challenges in real-time plagiarism detection," *Journal of Information Retrieval Research*, 2024.
- [6] H. Maurer, F. Kappe, and B. Zaka, "Plagiarism – a survey," *Journal of Universal Computer Science*, vol. 12, no. 8, pp. 1050–1084, 2006. doi: 10.3217/jucs-012-08-1050
- [7] Y. Zhao, L. Huang, and X. Li, "Syntactic analysis-based plagiarism detection," *Language Resources and Evaluation*, vol. 49, no. 3, pp. 555–582, 2015.
- [8] A. Barrón-Cedeño, P. Rosso, and E. Agirre, "Combining citation-based analysis with NLP for plagiarism detection," *Computational Linguistics and Digital Forensics*, 2013.
- [9] G. Pataki, "A fuzzy-semantic similarity approach for plagiarism detection," *International Journal of Computational Intelligence and Applications*, vol. 11, no. 4, pp. 1–12, 2012.
- [10] Y. Rajanak, R. Patil, and Y. P. Singh, "Language detection using natural language processing," in *Proc. 9th Int. Conf. on Advanced Computing and Communication Systems (ICACCS)*, Coimbatore, India, 2023, pp. 673–678. doi: 10.1109/ICACCS57279.2023.10112773
- [11] H. Shivnani, R. Sontakke, B. Sisode, P. Puri, and R. Patil, "Next word prediction system using NLP," in *Proc. Int. Conf. on Electrical Electronics and Computing Technologies (ICEECT)*, Greater Noida, India, 2024, pp. 1–5. doi: 10.1109/ICEECT61758.2024.10739289